Chapter 7 — Custom Hooks and the Rules of Hooks

You have used `useState` (Chapter 3), `useActionState` (Chapter 4), `useEffect` and `useRef` (Chapter 5), `useReducer` and `useContext` (Chapter 6). This chapter explains the **Rules of Hooks** properly — what they are, *why* React needs them, what breaks when ignored — then shows how to write your **own** Hooks: reusable **stateful logic** (code that remembers values between renders) such as `useLocalStorage`, `useDebounce` and `useFetch`. It finishes with built-ins that are easiest to grasp once you think in Hooks: `useId`, `useSyncExternalStore`, `useTransition`, `useDeferredValue` (plus `useEffectEvent` for callback-taking Hooks).

Code: `code/` — React 19.3 in jsdom with `@testing-library/react`. Fetching examples start a tiny `node:http` server on `127.0.0.1` (localhost) inside the script, so no internet is needed.

7.1 What a Hook really is

What it is

A **Hook** is a function whose name starts with `use` that lets a component "hook into" a React feature: state (`useState`), **side effects** (`useEffect`), **context** (`useContext`), and so on. Hooks only work **while React is rendering a component** — the only time React knows which component the call belongs to.

How it works — state is matched by call order

A component's local variables vanish when it returns, so React stores Hook state *outside* your function, in an internal record per component instance (a **fiber**) holding a **list of Hook slots**:

1. Before calling your component, React points a **cursor** at slot 0.
2. Each Hook call reads the current slot and advances the cursor.
3. The **only** thing identifying "this `useState` is the `name` one" is its **position**. Hooks have no names or keys.

A 15-line toy model in plain JavaScript reproduces this.

`code/01_hooks_by_call_order.js`:

```
let slots = [];      // this component instance's remembered values
let index = 0;       // which slot the next useState call reads

function useState(initial) {
  const i = index++;                      // position = identity
  if (!(i in slots)) slots[i] = initial;   // first render: store the initial value
  const setState = (v) => { slots[i] = v; };
  return [slots[i], setState];
}
function render(Component, props) { index = 0; return Component(props); }

// A "component" that breaks the rule: it calls a Hook only when a condition is true.
function Profile({ showNickname }) {
  const [name] = useState('Ada');
  if (showNickname) {
    const [nick] = useState('Countess');   // slot 1 ONLY on some renders
    void nick;
  }
  const [age] = useState(36);              // slot 1 or 2 depending on the condition!
  return `name=${name} age=${age}`;
}

console.log('render 1 (showNickname=true): ', render(Profile, { showNickname: true }));
console.log('slots now:', JSON.stringify(slots));
console.log('render 2 (showNickname=false):', render(Profile, { showNickname: false }));
```

Output:

```
render 1 (showNickname=true):  name=Ada age=36
slots now: ["Ada","Countess",36]
render 2 (showNickname=false): name=Ada age=Countess
-> age read slot 1, which holds the nickname. State got mixed up.
```

The skipped conditional call made `useState(36)` the *second* call, reading the nickname's slot — the whole reason for the Rules of Hooks.

Why React chose call order

Names or keys were considered. Position won because it keeps Hooks cheap (no string keys), makes custom Hooks **composable** (two custom Hooks can each call `useState` without name clashes) and allows the same Hook several times in one component. The price: *call Hooks in the same order every render*.

> **TL;DR**
>
> React stores Hook state in a per-component list and matches calls by **position**. Change the number or order of calls between renders and values land in the wrong slots.

7.2 The Rules of Hooks

> **What it is**

From react.dev:

1. **Only call Hooks at the top level** of a function component or custom Hook — never inside conditions, loops or nested functions.
2. **Only call Hooks from React functions** — components and custom Hooks, not plain JavaScript functions or class components.

The allowed and forbidden places

✅ Allowed	🔴 Not allowed
Top level of a function component body	Inside `if`, `&&`, `? :`, `switch` branches
Top level of a custom Hook body	Inside `for` / `while` loops
	After an early `return` (the Hook is skipped on some renders)
	Inside event handlers (`onClick={() => useState()}`)
	Inside the callbacks you pass to `useMemo`, `useReducer`, `useEffect`
	Inside `try` / `catch` / `finally` blocks
	In class components, or in plain helper functions

One exception: the `use` API (§9.11 of Chapter 9) *may* be called inside `if` and loops because it keeps no slot — but still not inside `try/catch`.

What React does when you break rule 1

`code/02_rules_of_hooks_errors.jsx`:

```jsx
// BAD: a Hook after an early return
function Greeting({ user }) {
  const [lang] = useState('en');
  if (!user) return <p>Please sign in ({lang})</p>;
  const [greeting] = useState('Hello');     // skipped when user is missing
  return <p>{greeting}, {user}</p>;
}

// BAD: a Hook inside a condition
function Tracker({ tracking }) {
  const [clicks] = useState(0);
  if (tracking) useEffect(() => {});
  return <p>clicks: {clicks}</p>;
}
// ... tryRender(label, firstUi, nextUi) renders, re-renders, prints the error ...

// GOOD: call every Hook unconditionally at the top, put the condition inside.
function GreetingFixed({ user }) {
  const [greeting] = useState('Hello');
  if (!user) return <p>Please sign in</p>;
  return <p>{greeting}, {user}</p>;
}
function TrackerFixed({ tracking }) {
  const [clicks] = useState(0);
  useEffect(() => { if (!tracking) return; /* start tracking */ }, [tracking]);
  return <p>clicks: {clicks}</p>;
}
```

Output:

```
  [dev warning] React has detected a change in the order of Hooks called by Greeting. This will lead to bugs and errors if not fixed. For more information, read the Rules of Hooks: https://react.dev/link/rules-of-hooks
Greeting: signed-out then signed-in -> Rendered more hooks than during the previous render.
Greeting: signed-in then signed-out -> Rendered fewer hooks than expected. This may be caused by an accidental early return statement.
  [dev warning] React has detected a change in the order of Hooks called by Tracker. This will lead to bugs and errors if not fixed. For more information, read the Rules of Hooks: https://react.dev/link/rules-of-hooks
Tracker : off then on               -> Rendered more hooks than during the previous render.
GreetingFixed     -> Hello, Ada
TrackerFixed      -> clicks: 0
```

A changed count makes React **throw** ("Rendered more/fewer hooks…"), crashing that part of the tree. If the count is the same but the *kind* of Hook changes, development builds warn about "a change in the order of Hooks" and you get §7.1's mixed-up state. The fix is always: **call the Hook unconditionally, move the condition inside it** (Effect body, returned JSX, or arguments).

Why rule 2 exists

A Hook needs a "currently rendering component" for its slot. A plain function called from a handler, a timer or **module scope** (top level of a file) has none → *"Invalid hook call. Hooks can only be called inside of the body of a function component."* The same message appears when the app **bundles two copies of React** (e.g. a linked library with its own `react`) — a classic **monorepo** (one repository holding several packages) bug.

The linter: `eslint-plugin-react-hooks`

Never find these bugs at **runtime**: the official **ESLint** plugin (a **linter** reads code without running it and flags likely bugs) checks as you type:

Rule	What it catches
`react-hooks/rules-of-hooks`	Hooks in conditions, loops, after early returns, in non-`use` functions
`react-hooks/exhaustive-deps`	Missing or unnecessary values in `useEffect` / `useMemo` / `useCallback` dependency arrays (§5.5)
Compiler-based rules (purity, refs, set-state-in-effect, …)	Code the React Compiler (auto-memoising build tool) cannot safely optimise — in the plugin's recommended preset in recent versions

Setup with ESLint 9's flat config (one `eslint.config.js` exporting an array; not run — Vite's React template already includes it):

```
// eslint.config.js
import reactHooks from 'eslint-plugin-react-hooks';
import { defineConfig } from 'eslint/config';

export default defineConfig([
  reactHooks.configs.flat.recommended,
]);
```

The linter recognises Hooks **by name**: `useSomething` (capital after `use`) is checked as a Hook; a capitalised function is treated as a component. Hence naming matters (§7.3).

Common mistakes

Mistake	Fix
`if (!data) return <Spinner/>;` above a `useMemo`	Move all Hooks above the first `return`
Calling a Hook in a `.map()` to make one per item	Extract an `<Item>` component and call the Hook inside it
Calling `useContext` in an event handler "to get the latest value"	Read it at top level; the handler closes over it
Silencing `exhaustive-deps` with `// eslint-disable-next-line`	Almost always hides a stale-value bug; restructure (Chapter 5, §7.7)
Naming a Hook `getUser()`	Call it `useUser()` so the linter checks it and its callers

In the interview

> **30-second answer:** "Two rules: call Hooks only at the top level of components or custom Hooks, and only from React functions. React stores Hook state in a list per component and matches calls by order, so the order must be identical every render — a Hook inside an `if` or after an early return shifts every later slot. React throws 'Rendered more/fewer hooks' when the count changes, and `eslint-plugin-react-hooks` catches it **statically** (by reading the code, before it runs)."

- *Why can `use` be conditional but `useState` can't?* — `use` stores nothing in a slot; it reads a promise or context each time.
- *Conditionally run an effect?* — Always call `useEffect`; put the `if` inside and include the condition in dependencies.
- *Is `const [a] = cond ? useState(1) : [0]` allowed?* — No — the Hook is conditional.
- *What else causes "Invalid hook call"?* — Calling a Hook outside a component, or two copies of React in the bundle.

TL;DR

Top level only, React functions only. The rule protects call order; the linter enforces it.

7.3 Custom Hooks — reusing stateful logic

What it is

A **custom Hook** is a function whose name starts with `use` and that **calls other Hooks**, extracting stateful logic for reuse — like a helper function, but for logic needing state or effects. Callers share the logic, each with its own state.

The problem it solves

Before Hooks, sharing stateful logic needed **higher-order components** (HOCs: functions taking a component and returning a wrapped one with extra props) and **render props** (a function prop that returns what to render) (Chapter 14), which nested the tree deeply ("wrapper hell"). A custom Hook is a plain call: no extra components or nesting, and you can call several.

How it works

1. Spot duplicated Hook logic (a toggle, a subscription, a fetch) and move it into `function useSomething(args) { … }`.
2. Return what callers need — a value, a **tuple** (fixed-length array with meaningful positions) `[value, setter]`, or an object `{ data, error, status }`.
3. Every call site gets its **own** state: `useToggle()` in two components runs `useState` twice, in two fibers.
4. The Hook re-runs on each render of its caller, so it always sees the latest arguments.

`code/03_custom_hook_basics.jsx`:

```jsx
function useToggle(initial = false) {
  const [on, setOn] = useState(initial);
  const toggle = () => setOn(v => !v);
  return [on, toggle];                    // return whatever shape is convenient
}

function Spoiler({ title, children }) {
  const [open, toggleOpen] = useToggle();
  return (
    <section>
      <button onClick={toggleOpen}>{open ? 'Hide' : 'Show'} {title}</button>
      {open && <p>{children}</p>}
    </section>
  );
}

function DarkSwitch() {
  const [dark, toggleDark] = useToggle(true);
  return <button onClick={toggleDark}>theme: {dark ? 'dark' : 'light'}</button>;
}
// render two <Spoiler>s and a <DarkSwitch>; click "Show ending", then "theme"

// Not every helper is a Hook: this one calls no Hooks, so it should NOT start with "use".
function formatCount(n) { return n === 1 ? '1 item' : `${n} items`; }
```

Output:

```
start          : Show ending | Show twist | theme: dark
open "ending"  : Hide ending | The butler did it. | Show twist | theme: dark
toggle theme   : Hide ending | The butler did it. | Show twist | theme: light
plain helper   : 3 items
```

Opening one spoiler didn't open the other; the theme switch has its own value: **custom Hooks share logic, never state**. For shared state, lift it, use context, or an external store (Chapter 6, §7.9).

Designing a good custom Hook

Guideline	Why
Name it `use` + CapitalLetter (`useOnlineStatus`)	The linter and React Compiler treat it as a Hook; readers know it may hold state
If it calls **no** Hooks, don't prefix it with `use`	It's a normal function; the prefix would wrongly restrict where it can be called
Name it after its **purpose**, not its mechanism	`useChatRoom(roomId)` beats `useEffectWithWebSocket`
Mirror built-in shapes where it helps	`useLocalStorage` returning `[value, setValue]` feels like `useState`
Return objects when there are 3+ values	`{ data, error, status }` is self-documenting and order-free
Keep it pure during render	Same rule as components: side effects go in Effects or handlers
Avoid "lifecycle" Hooks like `useMount(fn)`	They hide dependencies from the linter; write the Effect you mean
Add `useDebugValue(label)` for library Hooks	Shows a label next to the Hook in React DevTools (used in `code/hooks.js`)

Don't write one when the logic is short and used once (inline is clearer), when it is **server data** (fetch + cache + refetch — TanStack Query does it better, Chapter 9), or when you only share a *value* (props or context). **Testing:** through a tiny component (as here) or `renderHook` from `@testing-library/react` (Chapter 12).

In the interview

> **30-second answer:** "A custom Hook is a function starting with `use` that calls other Hooks. It's how you reuse stateful logic — like a subscription or a debounced value — without wrapper components. Each call has independent state, it re-runs on every render so it sees fresh arguments, and it must follow the Rules of Hooks. The `use` prefix is what lets the linter check it."

- *Do two components using the same custom Hook share state?* — No; share via lifting, context, or a store.
- *Custom Hooks vs HOCs/render props?* — Hooks compose without tree nesting and make data flow explicit; HOCs and render props remain valid but are rarely needed.

TL;DR

`useX` = a function that calls Hooks. It shares logic, not state. Name by purpose; don't prefix non-Hooks with `use`.

7.4 `useLocalStorage` — state that survives a reload

What it is

A Hook with `useState`'s API whose value is saved in `localStorage` (a small per-site key–value store keeping strings across reloads and restarts) and read back next visit.

How it works

1. **Lazy initial state**: pass a function to `useState` so storage is read once, on mount.
2. `JSON.parse` / `JSON.stringify` (**JSON** = JavaScript Object Notation, the standard text data format) because storage holds only strings.
3. `try/catch` around every storage call: it can throw (Safari private mode, **quota** — size limit — full, disabled cookies) and stored JSON can be corrupted.

4. An Effect writes the value when it changes — writing is a side effect, not render work.
5. A second Effect listens for the `storage` **event**, fired in *other* tabs of the same site when one tab changes a key — keeping tabs in sync.

`code/04_use_local_storage.jsx`:

```jsx
export function useLocalStorage(key, initialValue) {
  // 1. Lazy initial state: read storage once, on the first render only.
  const [value, setValue] = useState(() => {
    try {
      const raw = window.localStorage.getItem(key);
      return raw !== null ? JSON.parse(raw) : initialValue;
    } catch {
      return initialValue;                  // corrupted JSON or storage blocked
    }
  });

  // 2. Write whenever the value (or key) changes.
  useEffect(() => {
    try { window.localStorage.setItem(key, JSON.stringify(value)); } catch { /* quota / private mode */ }
  }, [key, value]);

  // 3. Other tabs fire a "storage" event when they change the same key.
  useEffect(() => {
    function onStorage(e) {
      if (e.key === key && e.newValue !== null) setValue(JSON.parse(e.newValue));
    }
    window.addEventListener('storage', onStorage);
    return () => window.removeEventListener('storage', onStorage);
  }, [key]);

  return [value, setValue];                 // same shape as useState, so updaters work too
}

function Settings() {
  const [fontSize, setFontSize] = useLocalStorage('fontSize', 16);
  return (
    <div>
      <p>font size: {fontSize}px</p>
      <button onClick={() => setFontSize(s => s + 2)}>bigger</button>
    </div>
  );
}
```

Output:

```
first visit     : font size: 16px
after 2 clicks  : font size: 20px | stored = 20
after reload    : font size: 20px
other tab -> 24 : font size: 24px
corrupted JSON  : font size: 16px (fell back to the default)
```

Trade-offs and limits

Concern	Note
Server rendering	No `localStorage` on the server. Reading it during render makes server HTML (default) differ from the client (stored value) → **hydration mismatch** (React warns and redraws). With SSR, render the default first and load the stored value in an Effect, or use `useSyncExternalStore` with `getServerSnapshot` (§7.9).
Two components, same key, same tab	Each has its own `useState` and doesn't see the other's changes (`storage` fires only in *other* tabs). If that matters, make storage an external store read with `useSyncExternalStore`.
Size	About 5 MB per **origin** (protocol + domain + port, e.g. `https://shop.example`); strings only; synchronous (blocks the **main thread** that runs JS and draws the page). Large data → IndexedDB (asynchronous browser database).
Security	Readable by any script on the page — never store tokens (session/access credentials) or personal data (see the JS book's storage chapter).

TL;DR

Lazy-read once, write in an Effect, `try/catch` everything, listen to `storage` for other tabs. Beware SSR mismatches.

7.5 `useDebounce` — wait until the user stops

What it is

Debouncing delays an action until input has been quiet for a set time: a search box firing per keystroke sends 5 requests for "react"; debounced by 300 ms it sends 1, after the pause.

How it works

1. Keep a second state, `debounced`, starting equal to `value`.
2. Each time `value` changes, an Effect schedules `setDebounced(value)` after `delayMs`.
3. The Effect's **cleanup** cancels the previous timer, so each keystroke cancels the last pending update — only the final value survives the pause.

`code/05_use_debounce.jsx`:

```jsx
export function useDebounce(value, delayMs = 300) {
  const [debounced, setDebounced] = useState(value);
  useEffect(() => {
    const id = setTimeout(() => setDebounced(value), delayMs);  // schedule
    return () => clearTimeout(id);           // a newer value cancels the pending one
  }, [value, delayMs]);
  return debounced;
}

function Search() {
  const [text, setText] = useState('');
  const query = useDebounce(text, 100);
  useEffect(() => { if (query) console.log(`  -> would search for "${query}"`); }, [query]);
  return <input aria-label="search" value={text} onChange={e => setText(e.target.value)} />;
}
// types r, re, rea, reac, react (one key every 30 ms), pauses, then types "react hooks"
```

Output:

```
typed "r"
typed "re"
typed "rea"
typed "reac"
typed "react"
(pause 150 ms)
  -> would search for "react"
typed "react hooks"
(pause 150 ms)
  -> would search for "react hooks"
```

Five keystrokes, one search. The input itself is **not** debounced — `text` updates immediately so typing feels instant; only the *derived* `query` waits.

Debounce vs throttle vs `useDeferredValue`

Technique	Behaviour	Use for
Debounce	Fire once after N ms of silence	Network requests while typing, autosave
Throttle	Fire at most once every N ms	Scroll/resize handlers, analytics
`useDeferredValue` (§7.11)	Re-render a slow part as soon as React has time; no fixed delay, interruptible	Expensive *rendering* (filtering a big list) — not network calls

`useDebouncedCallback(fn, ms)` debounces a *function* instead (autosave on change); libraries such as `use-debounce` provide both.

Common mistakes

- No cleanup → every keystroke eventually fires.
- Debouncing the controlled input's own value → the **caret** (text cursor) lags and characters appear late.
- Creating the debounced function inline each render (`debounce(fn, 300)` in the body) → a new timer per render, nothing debounced. Keep it in a ref or use the value-based Hook.

TL;DR

Timer in an Effect, cleared by the cleanup. Debounce the derived value, never the input.

7.6 `useFetch` — loading, error, success, and no races

What it is

A Hook that fetches a URL and returns `{ status, data, error }`, packaging the Effect-based fetching pattern and its **race-condition** fix from §5.7: when the URL changes while a request is in flight, the last response to arrive would otherwise win, even for the old URL.

`code/06_use_fetch.jsx` — the local server answers user 1 slowly (150 ms) and user 2 quickly (20 ms):

```jsx
// The naive version: no cleanup, so whichever response arrives LAST wins.
function useFetchNaive(url) {
  const [data, setData] = useState(null);
  useEffect(() => { fetch(url).then(r => r.json()).then(setData); }, [url]);
  return data;
}

export function useFetch(url) {
  const [state, setState] = useState({ status: 'idle', data: null, error: null });
  useEffect(() => {
    if (!url) return;
    const controller = new AbortController();          // lets cleanup cancel the request
    setState(s => ({ ...s, status: 'loading', error: null }));
    fetch(url, { signal: controller.signal })
      .then(res => { if (!res.ok) throw new Error(`HTTP ${res.status}`); return res.json(); })
      .then(data => setState({ status: 'success', data, error: null }))
      .catch(error => {
        if (error.name === 'AbortError') return;         // we cancelled it: not a real error
        setState({ status: 'error', data: null, error });
      });
    return () => controller.abort();                    // url changed or unmounted
  }, [url]);
  return state;
}

function Profile({ id }) {
  const { status, data, error } = useFetch(`${BASE}/api/users/${id}`);
  if (status === 'loading' && !data) return <p>Loading…</p>;
  if (status === 'error') return <p role="alert">Could not load user {id}: {error.message}</p>;
  if (!data) return null;
  return <p>User {id}: {data.name}{status === 'loading' ? ' (refreshing…)' : ''}</p>;
}
// render id=1, immediately switch to id=2, wait; then id=99 (does not exist)
```

Output:

```
naive         : Asked for user 2, showing: Ada <- WRONG user (race condition)
initial       : Loading…
switched to 2 : User 2: Grace
200 ms later  : User 2: Grace (the slow Ada response was aborted, not shown)
missing user  : Could not load user 99: HTTP 404
```

How it works

1. Each Effect run creates an `AbortController`; its `signal` goes to `fetch`. When `url` changes (or on unmount), cleanup calls `abort()` → the old `fetch` rejects with `AbortError`, which is ignored. (An `ignore` flag also prevents the stale `setState` but lets the request finish; aborting saves bandwidth.)
2. `fetch` rejects only on *network* failure, so `res.ok` is checked to turn 4xx/5xx **status codes** (e.g. 404 Not Found, 500 server error) into errors.
3. One state object with a `status` field makes impossible combinations (loading *and* error) impossible.
4. Keeping old `data` while `status === 'loading'` shows "refreshing…" instead of flashing a spinner.

When not to use it: fine as an exercise or one-off widget, but real apps need **caching**, **deduplication**, **refetch on focus**, **retries**, **pagination** and **mutations with invalidation** — exactly what TanStack Query or a router loader provides (Chapter 9). Effects also start fetching only *after* render, causing request **waterfalls** in nested components.

In the interview

30-second answer: "I'd keep `{status, data, error}` in one state object, start the fetch in an Effect keyed on the URL, pass an AbortController signal, and abort in the cleanup so a stale response can't overwrite a newer one. I'd check `res.ok` because fetch doesn't reject on 404. For production I'd use TanStack Query or router loaders to get caching, dedupe and retries."

TL;DR

Fetch in an Effect, abort in cleanup, check `res.ok`, one status field. For real apps, reach for a data library.

7.7 Hooks that take callbacks — `useEffectEvent`

What it is

Some custom Hooks accept a function — `useInterval(callback, ms)`, `useEventListener('keydown', handler)`, `useWebSocket(url, onMessage)`. The inner Effect must call the **latest** callback (seeing the latest props/state) without **restarting** the timer or connection each time the parent passes a new function.

`useEffectEvent` (stable since React 19.2) wraps a function into an **Effect Event**: a stable function that, called from an Effect, always runs the newest callback version. It is *not* reactive, so it stays out of the dependency array.

The problem it solves

Before, both options were bad: `callback` in the dependencies → the interval is torn down and recreated every render; leave it out (`[]`) → the Effect keeps the **first** callback forever (a **stale closure** — a function holding variables from an old render) and the linter complains.

`code/11_use_effect_event.jsx`:

```jsx
export function useInterval(callback, delayMs) {
  const onTick = useEffectEvent(callback);        // always calls the newest callback
  useEffect(() => {
    if (delayMs === null) return;                  // null = paused
    timersStarted++;
    const id = setInterval(() => onTick(), delayMs);
    return () => clearInterval(id);
  }, [delayMs]);                                   // onTick is deliberately NOT a dependency
}

// The tempting-but-wrong version: an empty dependency array freezes the FIRST callback.
function useIntervalStale(callback, delayMs) {
  useEffect(() => {
    const id = setInterval(() => callback(), delayMs);
    return () => clearInterval(id);
  }, []);                                          // the linter would warn: missing 'callback', 'delayMs'
}

// Stopwatch: useIntervalImpl(() => stepsSeen.add(step), running ? 20 : null)
// wait 70 ms, click "step" twice (step becomes 3), wait 70 ms
```

Output:

```
useInterval (useEffectEvent): ticks used step values 1, 3
useIntervalStale ([] deps)  : ticks used step values 1
intervals started: 2 (mount + resume; changing step never restarted it)
```

The correct Hook saw `step = 3` without restarting; the stale one kept `1`. Changing `delayMs` (pause → `null`, resume → `20`) *does* restart it, because the delay is a real dependency.

Rules (full list in §5.5): call Effect Events only from inside Effects (not during render or from handlers, which already see latest values); don't pass them to other components or Hooks; never list them as dependencies; use them for the *non-reactive* part of an Effect, not to hide real dependencies.

Before 19.2 the "latest ref" pattern did this — `const saved = useRef(callback); useEffect(() => { saved.current = callback; });` then `saved.current()` in the interval — still common in older code and libraries.

TL;DR

Callback-taking Hooks: wrap the callback in `useEffectEvent`, call it from the Effect, keep only truly reactive values in the dependencies.

7.8 `useId` — IDs for accessibility

What it is

`useId()` returns a **unique string ID** for the calling component instance, stable across re-renders and identical between server HTML and the hydrated client. It is for attributes linking elements by ID, including **ARIA** (`aria-*` attributes describing elements to screen readers): `<label htmlFor>`, `aria-describedby`, `aria-labelledby`, `aria-controls`.

The problem it solves

A reusable `<PasswordField>` needs an input `id` for its `<label>`. Hard-coded `id="password"` breaks when the field appears twice (duplicate IDs; the second label points at the first input). A global counter (`id = nextId++`) breaks under server rendering, where server and client may count in different orders → hydration mismatches.

`code/07_use_id.jsx`:

```
function PasswordField({ label }) {
  const id = useId();                       // one ID per component instance
  return (
    <div>
      <label htmlFor={`${id}-input`}>{label}</label>
      <input id={`${id}-input`} type="password" aria-describedby={`${id}-hint`} />
      <p id={`${id}-hint`}>At least 12 characters.</p>
    </div>
  );
}
function SignupForm() {
  return (<form><PasswordField label="Password" /><PasswordField label="Repeat password" /></form>);
}
// ... render, then renderToString + hydrateRoot, then an identifierPrefix
```

Output:

```
input id=_r_0_-input  aria-describedby=_r_0_-hint
input id=_r_1_-input  aria-describedby=_r_1_-hint
label lookup finds the 2nd input: true
its description: _r_1_-hint
server ids   : _R_1_-input, _R_1_-hint, _R_2_-input, _R_2_-hint
after hydrate: _R_1_-input, _R_1_-hint, _R_2_-input, _R_2_-hint
with identifierPrefix: _widget-R_0_-input
```

How it works

- The ID derives from the component's **position in the tree** (parent path), not a counter, so server and client agree when they render the same tree.
- Call it **once** per component; derive related IDs with suffixes (`${id}-input`, `${id}-hint`).
- `identifierPrefix` (option of `createRoot`, `hydrateRoot` and server renderers) stops IDs from two React **roots** on one page colliding.
- The format is an implementation detail that has changed between versions; never parse it or hard-code it in CSS or tests (query by role/label instead — which works *because* the IDs are wired correctly).

Not for list keys (keys come from data, `todo.id`; `useId` per item defeats the purpose), and not for database IDs or anything globally unique — use `crypto.randomUUID()` when *creating* data.

TL;DR

`useId()` → unique, SSR-safe ID per component for `htmlFor` / `aria-*`. One call, suffixes for related IDs. Never for keys.

7.9 `useSyncExternalStore` — subscribing to outside data

What it is

An **external store** is data living outside React state that can change on its own: browser APIs (`navigator.onLine` — online?, `matchMedia` — does a CSS media query match?, `window.innerWidth`, `localStorage`), third-party state libraries, a WebSocket cache. `useSyncExternalStore` lets a component **read** it and **re-render** on change, without the subtle bugs of `useState` + `useEffect`.

```
const snapshot = useSyncExternalStore(subscribe, getSnapshot, getServerSnapshot?);
```

Argument	Contract
`subscribe(callback)`	Start listening; call `callback` whenever the store changes; **return an unsubscribe function**. Should be the same function each render (define it outside the component) or React re-subscribes every render.
`getSnapshot()`	Return the current value. Must return the **same value (by `Object.is`)** if the store hasn't changed.
`getServerSnapshot()`	Optional: the value to use during server rendering and hydration. Required if the component renders on the server.

Why not `useState` + `useEffect`?

The Effect version subscribes *after* the first paint (the first render may be wrong), and under **concurrent rendering** (React can pause a render and continue later, §7.10) different components can read the store at different moments in one render and show inconsistent values — **tearing**. `useSyncExternalStore` reads snapshots in a way React can check for consistency, forcing a synchronous re-render if the store changes mid-render.

`code/08_use_sync_external_store.jsx`:

```
// ---- 1. A browser API: navigator.onLine + 'online'/'offline' events ----
function subscribeOnline(callback) {          // React passes a callback; call it on every change
  window.addEventListener('online', callback);
  window.addEventListener('offline', callback);
  return () => {                              // and return an unsubscribe function
    window.removeEventListener('online', callback);
    window.removeEventListener('offline', callback);
  };
}
function useOnlineStatus() {
  return useSyncExternalStore(
    subscribeOnline,                          // defined OUTSIDE the component: same function every render
    () => navigator.onLine,                   // getSnapshot (client)
    () => true                                // getServerSnapshot (used during SSR and hydration)
  );
}
function StatusBar() { return <p>{useOnlineStatus() ? 'Online' : 'Offline - changes will sync later'}</p>; }

// ---- 2. Your own tiny store (the pattern Redux/Zustand use internally) ----
function createStore(initial) {
  let state = initial;
  const listeners = new Set();
  return {
    getState: () => state,
    setState: (update) => { state = { ...state, ...update }; listeners.forEach(l => l()); },
    subscribe: (l) => { listeners.add(l); return () => listeners.delete(l); },
  };
}
const cart = createStore({ items: 0 });
function useCartCount() { return useSyncExternalStore(cart.subscribe, () => cart.getState().items); }

// ---- 3. The classic bug: getSnapshot returns a NEW object every time ----
function BadCart() {
  const snap = useSyncExternalStore(cart.subscribe, () => ({ ...cart.getState() })); // new object each call
  return <p>{snap.items}</p>;
}
```

Output:

```
start        : Online
wifi drops   : Offline - changes will sync later
wifi returns : Online
store update : header says 3 items | badge: 3
dev warning  : The result of getSnapshot should be cached to avoid an infinite loop
bad snapshot : Maximum update depth exceeded. This can happen when a component repeatedly calls setState inside
componentWillUpdate or componentDidUpdate. React limits the number of nested updates to prevent infinite loops.
```

How it works

1. On render React calls `getSnapshot()`; after commit it calls `subscribe(callback)`.
2. When the store changes, your code calls `callback`; React re-reads `getSnapshot()` and re-renders only if the value differs (`Object.is`).
3. On unmount (or if `subscribe` changes) React calls the unsubscribe function.

Part 3 shows why the snapshot must be *stable*: a fresh object is "different" every time → infinite re-renders. Return the same immutable object until it changes (as `cart.getState()` does), or select a primitive (`.items`).

Real world: Zustand and **React Redux** v8+ are built on it (Chapter 6); `useMediaQuery('(prefers-color-scheme: dark)')` subscribes to `matchMedia(...).addEventListener('change', …)` for theming (§8.8); routers subscribe to the history/location store this way.

When not to use it: if data can live in React state, keep it there. External store mutations can't be non-blocking **transitions** (§7.10) — a store update during a transition falls back to a blocking render — so prefer React state for UI that benefits from `useTransition`.

In the interview

> **30-second answer:** "`useSyncExternalStore(subscribe, getSnapshot, getServerSnapshot)` is the official way to read mutable data from outside React. React subscribes after commit and re-reads the snapshot on each change, re-rendering when it differs. It prevents tearing in concurrent rendering. The snapshot must be cached — return the same reference while nothing changed — and `subscribe` should be stable. Zustand and React Redux use it internally."

- *What is tearing?* — Different parts of the UI showing different versions of the same store during one render.
- `getSnapshot` *returns a new array each time?* — Infinite re-render loop; React warns "should be cached".

TL;DR

External data → `useSyncExternalStore`. Stable `subscribe`, cached `getSnapshot`, `getServerSnapshot` for SSR.

7.10 `useTransition` — non-urgent updates

What it is

React 18 introduced **concurrent rendering**: React can prepare new UI in the background, pause it for something more urgent, and discard it if outdated. You opt in by marking updates as **transitions**:

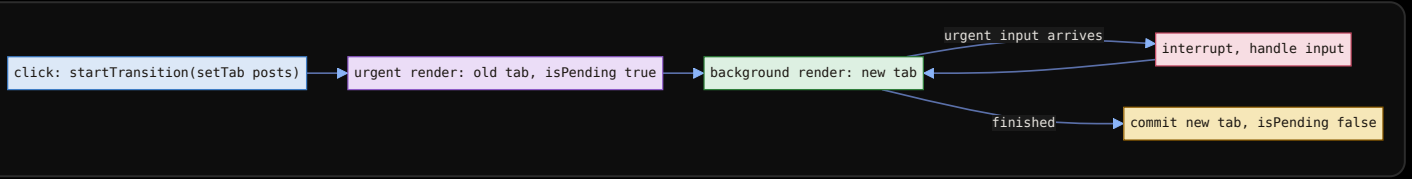
- **Urgent updates**: direct responses to input (typing, clicking) — expected instantly.
- **Transition updates**: moving between views (switching tabs, showing results) — a short delay is fine; freezing is not.

```
const [isPending, startTransition] = useTransition();
startTransition(() => setTab('posts'));          // this update is a transition
```

`isPending` is `true` while the transition is in progress — dim the old UI or show a small spinner instead of a blank screen.

How it works

1. `startTransition(fn)` runs `fn` immediately; updates inside it are tagged as transitions.
2. React first re-renders **urgently** with `isPending = true` and the *old* state — the current screen stays visible and interactive.
3. It renders the new state in the background; an urgent update arriving meanwhile interrupts it, is handled, and the transition restarts with the newest state.
4. When done, React commits and `isPending` becomes `false`.



`code/09_use_transition.jsx`:

```
function SlowPosts() {                              // pretend this tab is expensive to render
  const start = performance.now(); while (performance.now() - start < 30) { /* busy 30 ms */ }
  return <p>500 posts</p>;
}
const TABS = { about: () => <p>About me</p>, posts: SlowPosts };

function TabApp() {
  const [tab, setTab] = useState('about');
  const [isPending, startTransition] = useTransition();
  log.push(`render tab=${tab} isPending=${isPending}`);
  const Tab = TABS[tab];
  return (
    <div style={{ opacity: isPending ? 0.6 : 1 }}>
      <button onClick={() => startTransition(() => setTab('posts'))}>Posts</button>
      <Tab />
    </div>
  );
}

// React 19: the function passed to startTransition may be ASYNC (an "Action").
function RenameForm() {
  const [name, setName] = useState('ada');
  const [isPending, startTransition] = useTransition();
  seen.push(`${isPending ? 'Saving…' : 'Idle   '} name=${name}`);
  function handleSave() {
    startTransition(async () => {
      const saved = await saveToServer('grace');
      startTransition(() => setName(saved));      // updates AFTER an await need their own startTransition
    });
  }
  return <button onClick={handleSave}>Save</button>;
}
```

Output:

```
--- click "Posts" (update inside startTransition) ---
render tab=about isPending=true
render tab=posts isPending=false
--- async action ---
Idle    name=ada
Saving… name=ada
Saving… name=ada
Idle    name=GRACE
```

The first render after the click still shows **about** (dimmed, `isPending=true`); the expensive posts tab replaces it when ready. In the async example `isPending` stays true from click until the whole async function — network wait included — finishes and the new name applies (React may render more than once while pending).

React 19: async transitions are "Actions"

Since React 19, `startTransition` accepts an **async function**, called an **Action**: React tracks its pending state, and it's the mechanism behind form `action` props, `useActionState` and `useOptimistic` (Chapter 4). Current limitation: state updates **after an** `await` aren't automatically in the transition — wrap them in another `startTransition`, as above.

Transitions and Suspense

If a transition makes a component **suspend** (pause rendering because something isn't ready, e.g. route data — Chapter 9), React **keeps showing the old screen** instead of replacing visible content with a `<Suspense>` fallback (placeholder UI such as a spinner). That's why routers wrap navigations in transitions: no spinner flashes between pages.

Trade-offs and caveats

Point	Detail
Can't control text inputs	Updates to an input's `value` must be urgent; a transition would make the caret lag. Keep two states (urgent text + transitioned results) or use `useDeferredValue`.
The callback runs synchronously	`startTransition` doesn't delay anything like `setTimeout`; it only lowers the priority of the updates.
Not for making slow code fast	It keeps the UI *responsive* while slow rendering happens; to reduce the work itself, see Chapter 10.
No `isPending` needed?	Import the standalone `startTransition` from `react` (usable outside components, e.g. in a store).

In the interview

> **30-second answer:** "`useTransition` marks state updates as non-urgent. React renders them in the background and can interrupt them for urgent input, while `isPending` lets me show the old UI dimmed instead of freezing or flashing a fallback. In React 19 the callback can be async — an Action — and `isPending` covers the whole async operation. I can't use it for the value of a controlled input."

- *`useTransition` vs `useDeferredValue`?* — `useTransition` wraps the *update* (you own the setter); `useDeferredValue` wraps a *value* (useful when you only receive it, e.g. as a prop).
- *Is it a debounce?* — No: no timer; it runs as soon as possible and yields to urgent work.

TL;DR

`startTransition(() => setX(...))` = "update when you can, keep the old UI meanwhile". `isPending` for feedback. Async callbacks = Actions (React 19).

7.11 `useDeferredValue` — let part of the UI lag behind

What it is

`useDeferredValue(value)` returns a **deferred copy** of `value`: on an urgent update it first returns the **old** value (so fast parts update immediately), then schedules a background re-render with the new value — the same interruptible, low-priority render a transition uses.

```
const deferredQuery = useDeferredValue(query);                // React 18+
const deferredQuery = useDeferredValue(query, initialValue);  // React 19: value for the very first render
```

How it works

1. The user types; `text` updates urgently and the component renders with `text = "re"` but `deferred = ""`.
2. A `memo`-wrapped child (skips re-rendering when props are unchanged; Chapter 10) receiving `deferred` sees the **same prop** and is skipped — the input stays snappy.
3. React re-renders in the background with `deferred = "re"`; now the slow child renders.
4. If another key arrives first, React abandons that render and restarts with the newest value.

`code/10_use_deferred_value.jsx`:

```jsx
// memo is essential: it lets React skip SlowList when only the urgent `text` changed.
const SlowList = memo(function SlowList({ query }) {
  log.push(`  SlowList renders with query="${query}"`);
  const start = performance.now(); while (performance.now() - start < 20) { /* expensive */ }
  return <ul>{WORDS.filter(w => w.startsWith(query)).map(w => <li key={w}>{w}</li>)}</ul>;
});

function Search() {
  const [text, setText] = useState('');
  const deferred = useDeferredValue(text);           // lags behind `text` during typing
  const isStale = text !== deferred;
  log.push(`render text="${text}" deferred="${deferred}"${isStale ? '  (list is stale)' : ''}`);
  return (
    <>
      <input aria-label="q" value={text} onChange={e => setText(e.target.value)} />
      <div style={{ opacity: isStale ? 0.5 : 1 }}><SlowList query={deferred} /></div>
    </>
  );
}

// React 19: an initial value lets the FIRST render be cheap too.
function Preview({ text }) {
  const shown = useDeferredValue(text, '');          // first render uses '' then catches up
  log.push(`Preview render shown="${shown}"`);
  return <p>{shown}</p>;
}
```

Output:

```
render text="re" deferred=""  (list is stale)
render text="re" deferred="re"
  SlowList renders with query="re"
list now: react, redux, reducer, ref, render
Preview render shown=""
Preview render shown="hello"
```

The first render after the keystroke didn't run `SlowList` (memo saw the same `query`), so the input updated instantly; the list caught up in the background render. `text !== deferred` is a free "stale" flag for dimming.

Common mistakes

Mistake	Fix
No `memo` on the slow child	Without it the child re-renders on the urgent pass too, and nothing is gained
Passing a new object each render (`useDeferredValue({ q })`)	Defer primitives or memoised values; a new object is "new" every time
Using it to reduce network requests	It has no delay; it doesn't limit requests — debounce instead (§7.5)
Expecting it to help with slow *effects*	It only defers rendering

With Suspense: if the deferred value makes a child suspend (loading data for the new query), React keeps showing the **previous results** instead of a fallback — "stale-while-loading" search for free.

TL;DR

`useDeferredValue(v)` → urgent render with the old value, background render with the new one. Pair with `memo`. Not a debounce.

7.12 Putting it together — a user search

Four Hooks compose into one feature: `useId` wires the label, `useDebounce` limits requests, `useFetch` loads results without races, `useLocalStorage` remembers recent searches. Final Hook versions are in `code/hooks.js` (with a `useDebugValue` label on `useFetch`); the component is `code/UserSearch.jsx`:

```
export function UserSearch({ apiBase = '', initialQuery = '' }) {
  const id = useId();
  const [text, setText] = useState(initialQuery);
  const query = useDebounce(text.trim(), 250);
  const { status, data, error } = useFetch(query ? `${apiBase}/api/users?q=${encodeURIComponent(query)}` : null);
  const [recent, setRecent] = useLocalStorage('recent-searches', []);

  function remember(term) { setRecent(r => [term, ...r.filter(t => t !== term)].slice(0, 3)); }

  return (
    <div className="search">
      <label htmlFor={id}>Find a person</label>
      <input id={id} value={text} placeholder="Type a name…" onChange={e => setText(e.target.value)}
             onKeyDown={e => { if (e.key === 'Enter' && text.trim()) remember(text.trim()); }} />
      {recent.length > 0 && <p className="recent">Recent: {recent.join(', ')}</p>}
      {status === 'loading' && <p className="muted">Searching…</p>}
      {status === 'error' && <p role="alert">Search failed: {error.message}</p>}
      {status === 'success' && (data.length === 0
        ? <p className="muted">No one called “{query}”.</p>
        : <ul>{data.map(u => <li key={u.id}><strong>{u.name}</strong> <span>{u.role}</span></li>)}</ul>)}
    </div>
  );
}
```

`code/12_user_search.jsx` runs it against a local server that records every request:

```
typed a, ad, ada quickly -> requests sent: ["ada"]
results: Ada Lovelace Analyst
after "zed": No one called “zed”.
Recent: zed, ada | stored: ["zed","ada"]
```

In the browser (`code/search.html`; `fetch` is mocked in `search-main.jsx` so it works offline):

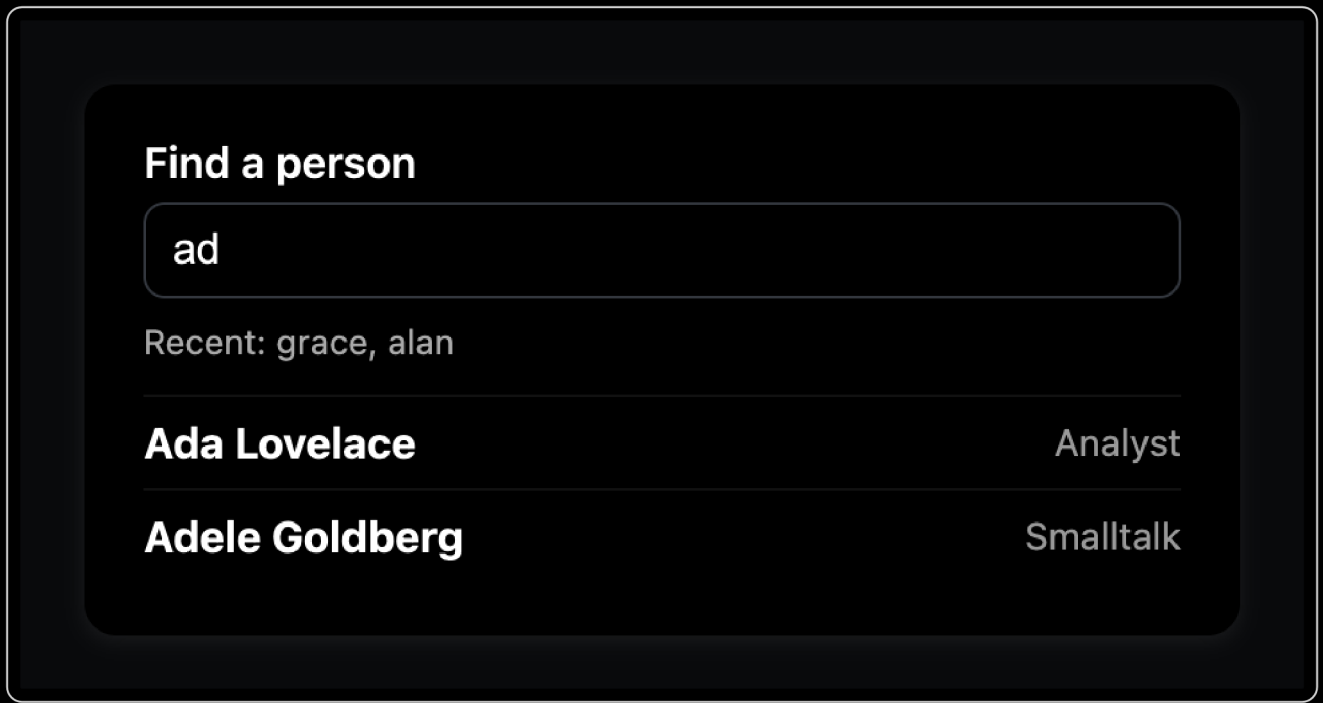


Figure 7.1 — `UserSearch` in Chrome. The component itself contains no timers, `fetch` calls or storage code — each concern is one Hook call.

Design notes: the component reads like a description of the feature, and each Hook is testable and reusable alone; swapping `useFetch` for TanStack Query's `useQuery` (Chapter 9) changes one line and adds caching; three typed characters made one request — debounce + abort keep the server quiet and results correct.

Chapter 7 TL;DR

Idea	Remember
How Hooks work	State lives in a per-component list, matched by **call order**
Rules of Hooks	Top level only (no `if`/loops/early-return/`try`/handlers); only in components and custom Hooks; enforced by `eslint-plugin-react-hooks`
Custom Hooks	`useX` functions that call Hooks; share **logic**, not state; name by purpose
`useLocalStorage`	Lazy read, write in Effect, `try/catch`, `storage` event; beware SSR mismatch
`useDebounce`	Timer in Effect + cleanup; debounce the derived value, not the input
`useFetch`	AbortController in cleanup, check `res.ok`, one status field — then prefer TanStack Query
`useEffectEvent`	Latest callback inside an Effect without re-subscribing (React 19.2)
`useId`	SSR-safe unique IDs for `htmlFor`/`aria-*`; never for keys
`useSyncExternalStore`	Read outside data; stable `subscribe`, cached `getSnapshot`, `getServerSnapshot`
`useTransition`	Mark updates non-urgent; `isPending`; async callbacks = Actions (19)
`useDeferredValue`	Deferred copy of a value; needs `memo`; `initialValue` in 19; not a debounce

Next: Chapter 8 — Styling.